CS766 Project Mid-term Report: Predicting Pedestrian Crossing Intention for Autonomous Vehicles

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1 Introduction

Urban pedestrian safety remains one of the most consequential open problems in autonomous driving. While perception systems have become adept at detecting pedestrians, anticipating what a pedestrian intends to do is a fundamentally harder and more important problem. A system that reacts only after a pedestrian steps into the road is already operating at the edge of its response window whereas genuine safety demands predicting crossing behavior before it happens. This has motivated a growing body of work that treats the problem as inferring a binary crossing decision from visual observations of the pedestrian and their surrounding scene. Yet despite strong benchmark performance, existing models are almost exclusively evaluated under clean, unobstructed conditions that rarely reflect the complexity of real urban environments. It remains to investigate how these systems behave when faced with partial occlusions, varying pedestrian appearances, and the visual noise inherent to dense streetscapes. This project investigates these questions directly, examining how state-of-the-art intent prediction models respond to systematically varied occlusion conditions and appearance perturbations, with the goal of better characterizing model robustness in the settings that matter most for real-world deployment.

2 Methodology

EfficientPIE [1] is a purely convolutional architecture with no recurrent components. It takes a single 300×300 RGB image cropped around the pedestrian bounding box as input and passes it through six blocks - two Common-Conv, two Fused-MBConv, and two MBConv, followed by global average pooling and a binary classifier. All blocks use depthwise separable convolutions, which reduce computational cost to approximately $\frac{1}{9}$ of standard convolutions, enabling 0.21 ms per-frame inference. The later MBConv blocks additionally incorporate squeeze-and-excitation [6] channel attention to improve representational capacity.

Training uses two techniques beyond standard cross-entropy optimization. *Intention domain incremental learning (IDIL)* trains the model progressively across temporal subsets of the data using an adaptive loss that penalizes catastrophic forgetting, accounting for a 3–4% accuracy gain in ablations. *Progressive perturbation* adds linearly increasing noise to output logits during backpropagation to exploit the inherent uncertainty in crossing intention labels and improve generalization. Together, these bring the model to 92% accuracy on PIE and 89% on JAAD.

We additionally explored PedVLM [2] as a complementary baseline, representing a contrasting paradigm to EfficientPIE's purely convolutional design. PedVLM is a vision-language model that combines a CLIP-ViT vision encoder (applied independently to both RGB images and optical flow) with a T5-based language model, fusing visual embeddings through gated attention pooling with text prompts encoding pedestrian attributes, trajectory history, and scene context to jointly predict crossing intention and generate natural language explanations. The model is designed to be trained on the PedPrompt dataset, which is built upon JAAD, PIE, and the TITAN dataset. However, due to the storage and disk space constraints discussed in 3.1, we were unable to include PIE and TITAN data in our training pipeline, restricting training to JAAD alone. This resulted in performance below the numbers reported in the original paper, and we therefore do not report PedVLM as a primary baseline. Notably, even in the original paper, PedVLM's performance degrades substantially on PIE relative to JAAD, which the authors attribute to frequent occlusions and distant pedestrians in that dataset, a finding that further motivates our focus on occlusion robustness in our revised timeline.

Furthermore, we explored STR-PIP[3], a spatiotemporal relationship reasoning framework that explicitly models physical interactions through a dynamic, pedestrian-centric scene graph. The architecture parses the scene using an off-the-shelf instance segmentation model to extract binary masks of contextual objects, isolates the target pedestrian with ground-truth bounding boxes, and encodes both modalities using separately tuned ResNet-18 backbones. For each frame, a star graph is constructed where edge weights between the central pedestrian node and peripheral object nodes are determined by relative spatial coordinates and visual appearance features. To capture temporal dynamics, Gated Recurrent Units (GRUs) connect the pedestrian nodes and context features across consecutive frames, ultimately passing the aggregated representations to a final prediction GRU to anticipate future crossing behavior.

3 Completed Work

3.1 Dataset Acquisition and Infrastructure

Our proposal planned to use both the PIE [4] and JAAD [5] datasets as twin evaluation benchmarks. During the dataset preparation phase, we encountered a significant practical constraint: the PIE dataset's raw video frames total in excess of 200 GB of storage. Given that our primary computational environment is Google Colab Pro, with its shared and ephemeral disk allocation, reliably hosting and preprocessing this volume of data proved infeasible within our resource envelope.

Rather than compromise the quality of our preprocessing pipeline or risk mid-experiment storage failures, we made a deliberate decision to focus our efforts on the JAAD dataset, which is substantially more compact while still being a well-established and widely cited benchmark for pedestrian intent prediction.

The JAAD dataset consists of 346 short video clips captured from a front-facing dashboard camera across a range of urban environments, lighting conditions, and pedestrian densities. It provides per-pedestrian bounding box annotations and crossing/not-crossing behavior labels, yielding 40,046 samples after clipping with an overlap ratio of 0.5.

3.2 Data Preprocessing Pipeline

A preprocessing pipeline was run to transform raw JAAD video and annotation data into model-ready inputs. The key steps were as follows:

- Video clips were decoded and frames extracted using the JAAD dataset API, generating pedestrian tracks with a sequence overlap rate of 0.5 and a maximum observation window of 15 frames, yielding a balanced set of crossing and non-crossing samples.

- For each sample, a context-aware crop of 2× the bounding box size was extracted around each labeled pedestrian to capture local scene context, then zero-padded and resized to 300×300 pixels using the pad-resize mode, consistent with the EfficientPIE input format.
- During training, random horizontal flipping and color jitter (brightness, contrast, saturation, and hue) were applied to improve generalization. All images were normalized with mean and standard deviation of 0.5 across all three channels.

3.3 EfficientPIE Baseline Reproduction

Following successful dataset preparation, we trained EfficientPIE from scratch on our preprocessed JAAD splits as the primary baseline for this project. The model was trained for 40 epochs using the RMSProp optimizer with a weight decay of 10^{-4}, a base learning rate of 10^{-5}, and a cosine annealing schedule with warm restarts. Training was conducted with a batch size of 32 on a Google Colab Pro GPU, with Google Drive mounted for persistent checkpoint storage across sessions.

4 Results

After 40 epochs of training on JAAD, the model was evaluated on the held-out test set (59 batches). Table 1 compares our results against the authors' reported JAAD figures from [1]. The curves for loss, accuracy, recall and F1-score are shown in Figures 1 to 4 respectively.

Table 1: Test Set Results vs. Authors' Reported JAAD Performance

	Loss	**Accuracy**	**Precision**	**Recall**	**F1**	**AUC**
Ours	0.368	0.874	0.592	0.620	0.605	0.856
EfficientPIE	—	0.890	0.630	—	0.620	0.860
Gap	—	*−1.6%*	*−3.8%*	—	*−1.5%*	*−0.4%*

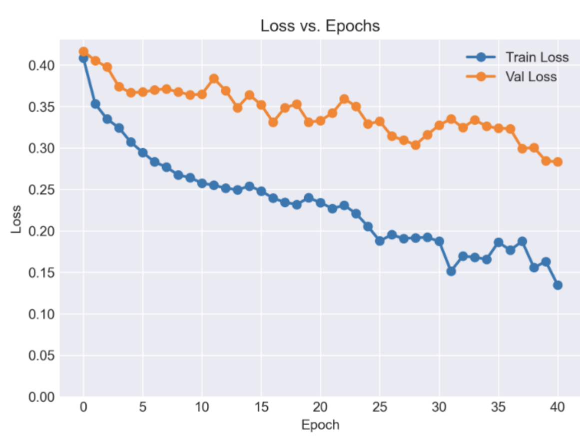

Figure 1: Training and Validation loss

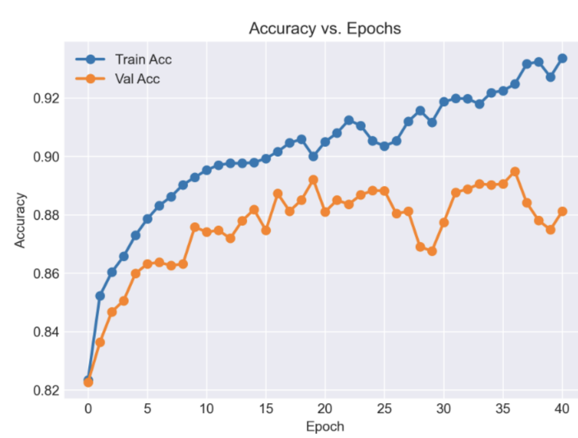

Figure 2: Training and Validation accuracy

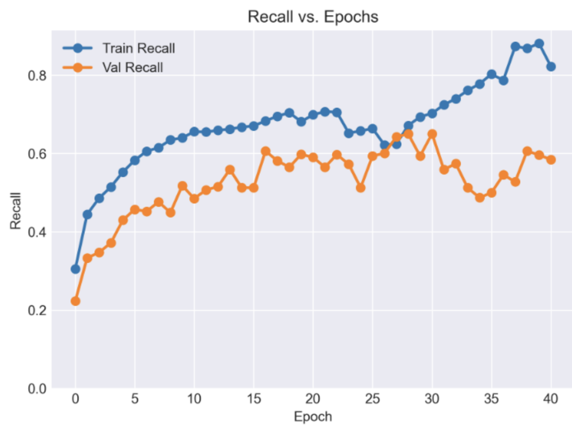

Figure 3: Training and Validation Recall

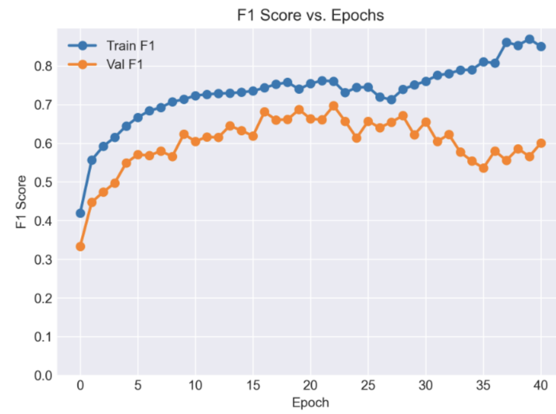

Figure 4: Training and Validation F1 score

5 Challenges

5.1 Compute

The primary constraint we faced was limited compute availability. Operating exclusively within Google Colab Pro, which imposes restrictions on GPU availability, session duration, and concurrent usage, meant that the total number of experiments we could run was tightly bounded. This made systematic hyperparameter tuning infeasible, as each configuration change would consume a significant portion of our available compute budget. Additionally, frequent Colab session disconnections required us to implement robust checkpointing to safeguard training progress, further adding to the overhead of each experimental run.

5.2 PIE Dataset Infeasibility

The PIE dataset's storage requirements - raw video, annotations, and extracted frames combined, exceeded both the available Google Drive quota and Colab RAM limits, making it impossible to train on PIE in the current setup. This was a significant deviation from the original proposal, which planned baseline reproduction on both datasets. Consequently, this infrastructure constraint affected all three baseline models we attempted.

6 Revised Timeline

Table 2: Revised Project Timeline

Weeks	Milestone
Weeks 1–4	*(Completed)* Environment setup, JAAD preprocessing, and EfficientPIE baseline training and evaluation.
Week 5–6	Occlusion robustness analysis and pedestrian scale analysis
Week 7	Consolidate findings, produce performance tables and visualizations.
Week 8	Final report write-up, project webpage, and submission.

References

[1] Fang Qu, Pengzhan Zhou, Yuepeng He, Kaixin Gao, Youyu Luo, Xin Feng, Yu Liu, and Songtao Guo. 2025. EfficientPIE: real-time prediction on pedestrian crossing intention with sole observation. *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence (IJCAI '25).*

[2] F. Munir, S. Azam, T. Mihaylova, V. Kyrki and T. P. Kucner, "Pedestrian Vision Language Model for Intentions Prediction," in *IEEE Open Journal of Intelligent Transportation Systems*, vol. 6, pp. 393-406, 2025, doi: 10.1109/OJITS.2025.3554387.

[3] B. Liu et al., "Spatiotemporal Relationship Reasoning for Pedestrian Intent Prediction," in *IEEE Robotics and Automation Letters*, vol. 5, no. 2, pp. 3485-3492, April 2020, doi: 10.1109/LRA.2020.2976305.

[4] A. Rasouli, I. Kotseruba, T. Kunic and J. Tsotsos, " PIE: A Large-Scale Dataset and Models for Pedestrian Intention Estimation and Trajectory Prediction," *2019 IEEE/CVF International Conference on Computer Vision (ICCV)*, Seoul, Korea (South), 2019, pp. 6261-6270, doi: 10.1109/ICCV.2019.00636.

[5] Kotseruba, I., Rasouli, A., & Tsotsos, J. K. (2016). Joint Attention in Autonomous Driving (JAAD). *arXiv preprint arXiv:1609.04741.*

[6] Hu, J., Shen, L., Albanie, S., Sun, G., & Wu, E. (2019). Squeeze-and-excitation networks. IEEE Transactions on Pattern Analysis and Machine Intelligence, 42(8), 2011–2023. doi.org.